

to the fact that given any two Freedonian senators, $\dots$
[$\therefore$ Either $x=0$ or $x=1$]. Since there is atleast one honest Freedonian senator,

[$\therefore x=1$] [Final Answer: 1 honest, 49 corrupt]

1.3 Propositional Equivalences

Definition: 1) A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**.
A compound proposition that is always false is called a **contradiction**. A compound proposition that is neither a tautology nor a contradiction is called a **contingency**.

Example 1

We can construct examples of tautologies and contradictions using just one propositional variable. Consider the truth tables of $(p \vee \neg p)$ and $(p \wedge \neg p)$, shown in Table 1. Because $p \vee \neg p$ is always true, it is a tautology. Because $p \wedge \neg p$ is always False, it is a contradiction.